



Armada Kalamunda Group

Pre-Labour Rupture of Membranes > 37 weeks Guideline

1. Purpose

This guideline provides guidance on the management options of Pre-labour Rupture of Membranes (PROM) from 37 weeks gestation in order to minimize its associated maternal and fetal risks.

2. Applicability

This guideline applies to medical and midwifery staff working within the Armadale Health Service maternity unit.

3. Background

To reduce the risks associated with prolonged PROM at term that relate to maternal and neonatal infection, cord prolapse and fetal compromise resulting in operative delivery or low five minute Apgar score.

- The incidence of Chorioamnionitis in women at term with pre labour rupture of the membranes is 6-10 % and occurs in up to 40 % of women with membrane rupture greater than 24 hours.
- The increasing numbers of digital vaginal examinations, longer duration of active labour and meconium staining of the amniotic fluid are the most important risk factors for the development of chorioamnionitis. Using this knowledge to counsel women about active management with early induction of labour at 24 hours with no GBS.
- To counsel active management and commencement of antibiotic therapy on confirmation of PROM in women with a positive GBS status.
- The overall risk of maternal postpartum endometritis is 3-4 %
- Refer to Appendix: Quick Reference Flowchart
- There is no universally accepted definition of PROM and hence failure to establish in labour after a latent period of 4 hours following confirmed rupture of membranes has been adopted
- PROM situations arise in 6 – 19 % of women at term.
- Even when the cervix is unfavourable, the majority of women labour spontaneously within 12 hours.
 - 50 % of these women will be in labour after 12 hours.
 - 86 % will be established in labour within 24 hours.
 - 94 % will be established in labour within 48 – 95 hours.
 - 6 % of women will not establish in labour within 96 hours of PROM

4. Management

4.1 Initial telephone consultation

- When a woman telephones with a history of suspected PROM, ask her to put a sanitary pad in place before presenting to Antenatal Assessment Unit (AAU).

4.2 Assessment

- Take history and perform general and obstetric examination observations.
- Document any previous obstetric history including history of Caesarean section.
- Observe underwear or sanitary wear for evidence of liquor.
- Listen to the fetal heart after abdominal examination.
- Clarify the woman's Group B Streptococcus (GBS) status.
- Offer the woman to empty her bladder prior to sterile speculum examination.
- NB: Sterile speculum examination avoids the need for digital vaginal examination reducing risk of infection.

All non-cephalic presentations require a digital examination to exclude cord prolapse.

4.3 Confirm diagnosis of PROM

- Obvious liquor on pad or underwear
- Pooling of liquor in the posterior fornix if no obvious loss seen on pad
- Amnio sure positive reaction: two pink lines
- Visually estimate cervical dilatation if possible
- Blood or meconium present
- Exclude cord prolapse.
- Take vaginal microbiological swabs (including GBS screening if results not available or not already taken)

4.4 Cardiotocography

- If PROM is confirmed, perform CTG to ensure fetal wellbeing.

4.5 Active or conservative management

- The woman and her partner need to be counselled about the management options of active or conservative management for PROM at term, and in relation to the woman's GBS status.
- Regardless of any other clinical factors, women at term who have rupture of the membranes for ≥ 18 hours should be commenced on parenteral antibiotic cover and induction of Labour (IOL) advised at 24 hours.

4.6 GBS negative

4.6.1 Expectant management

Women with term PROM who are Group B Streptococcus negative and choose expectant management of labour will be given a time for admission to commence antibiotic prophylactic

cover and induction of labour before PROM exceeds 24 hours with appropriate follow up of mother and baby.

If the woman chooses conservative management at home after proper counselling, then:

- She may go home if CTG is normal.
- Advise the woman to record her temperature every 4 hours and to observe vaginal loss, fetal movements and uterine contractions. Provide her with disposable thermometer to record temperature.
- Advise the woman to return to the hospital for admission and management plan if she develops a temperature ($> 37.5^{\circ}\text{C}$), the colour or the odour of vaginal loss changes, the baby does not move as much as previously, or contractions start.
- Women who have not established in labour after PROM exceeds 24 hours should be advised to return to AAU for induction of labour.

4.6.2 GBS unknown

- If the woman's GBS status is unknown she should be offered induction within 18 hours post PROM.

4.6.3 GBS positive, active management

- It is recommended that women who are carriers of Group B Streptococcus commence antibiotic therapy when PROM is confirmed and are induced within six hours of PROM.
- Refer to GBS guideline.

4.7 Induction/augmentation with syntocinon infusion

- If labour does not establish after a latent period of 4 hours from PROM.
- Prostaglandin Induction if Bishop Score below 6 or deemed unfavourable.
- Induction of labour (IOL) with vaginal prostaglandins is associated with an increased risk of chorioamnionitis and neonatal infection in comparison with an oxytocin induction.
- Oxytocin rather than vaginal prostaglandins is preferred for the IOL in the presence of PROM at term.
- If cervix favourable or Bishop Score greater than 6, commence Syntocinon infusion.

4.8 Intrapartum antibiotics for endometritis prophylaxis

Parenteral antibiotic cover is required in all cases (irrespective of GBS status) of PROM > 18 hours (GBS unknown) to 24 hours (GBS negative) until birth.

- Benzyl penicillin 3gram IV stat followed by Benzyl penicillin 1.8gm every 4 hours until birth.
- Clindamycin 900 mg IV 8 hourly until birth if allergic to penicillin.
- If the woman is at high risk of anaphylaxis, and the susceptibilities of the isolate is known to be resistant to Clindamycin; Vancomycin 20 mg/kg up to 1.5 grams 12 hourly until delivery.

4.9 Signs and symptoms of chorioamnionitis

Diagnosis relies on clinical presentation and / or laboratory investigations:

- Maternal fever $> 38^{\circ}\text{C}$ on two separate occasions* with any 2 of the following
- Maternal tachycardia ($> 100\text{ bpm}$)

- Fetal tachycardia (>160 bpm)
- Uterine tenderness
- Offensive smelling vaginal discharge
- Increased white cell count (>15µL)
- C-Reactive Protein > 40IU

(*With maternal fever, ensure well hydrated prior to next temperature recording)

4.9.1 Treatment of suspected/confirmed Chorioamnionitis

- There needs to be low tolerance of any suspicion of Chorioamnionitis.
- Commence Amoxicillin 2 G IV initial dose then 1g IV every 4 hours, Gentamycin as per AMS guidelines and Metronidazole 500 mg IV every eight hourly.
- Delivery of the fetus is indicated by immediate induction / augmentation or Caesarean section depending on the clinical situation and CTG.

4.10 Placenta and Membranes

- Histological examination of placenta and membranes with evidence of acute inflammation may confirm diagnosis post birth.
- Placental swab should be taken for microscopy, culture and sensitivity.

4.11 Neonatal care

- Inform the Paediatrician to be present at birth if fetal tachycardia and woman being treated for suspected Chorioamnionitis.
- Routine observations should be done for possible sepsis of the neonate as for GBS positive woman.

5. Legislative context

The following documents are required to give effect to this policy:

- Health Department of Western Australia (HDWA) [Clinical Handover Policy MP 0095](#)
- East Metropolitan Health Service (EMHS) [Fetal Surveillance and Cardiotocography \(CTG\) Policy EMHS: 195](#)

6. Supporting information

The following documents support the implementation of this policy:

- [Group B Streptococcus Guideline OBS-GUI-0184-16](#)
- [Antimicrobial Stewardship Policy AKG-POL-0039-18](#)
- [Medication Management Procedure AKG-PRO-0255-18](#)

7. Compliance, monitoring and evaluation

Compliance against this guideline will be monitored via:

- Monitoring of reported clinical incident data via Datix CIMS.

Department specific findings are to be reported at the departmental meeting for review and

identification of any opportunities for improvement.

8. Relevant standards

National Safety and Quality Health Service (NSQHS) Standards (second edition):

- Standard 3 – Action 3.15 Antimicrobial stewardship
- Standard 5 – Action 5.11 Clinical Assessment
- Standard 6 – Action 6.9 Communication critical information
- Standard 6 – Action 6.7 Clinical Handover

9. References

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2. King Edward Memorial Hospital. Rupture of Membranes – Spontaneous: Pre-labour rupture of membranes at term. Obstetrics & Gynaecology Clinical Practice Guideline; November 2019
3. . National Institute for Health & Care Excellence. Clinical guideline [CG190] 1.11. Prelabour rupture of membranes at term. February 2017. <https://www.nice.org.uk/>
4. Neil PR, Wallace EM. Is Amnisure(R) useful in the management of women with prelabour rupture of the membranes? The Australian & New Zealand journal of obstetrics & gynaecology. 2010;50(6):534-8.
1. Centers for Disease Control and Prevention. Prevention of Perinatal Group B Streptococcal Disease Revised Guidelines from CDC, 2010 Continuing Education. In: Services DoHaH, editor. 2010.
2. Flenady V, King J. Antibiotics for prelabour rupture of membranes at or near term. The Cochrane database of systematic reviews. 2002(3):CD001807.
3. Passos F, Cardoso K, Coelho AM, Graca A, Clode N, Mendes da Graca L. Antibiotic prophylaxis in premature rupture of membranes at term: a randomized controlled trial. Obstetrics and gynecology. 2012;120(5):1045-51.
4. Weekes CR, Mahomed K. Management of pre-labour rupture of membranes at term - a survey of current practice in Australia. The Australian & New Zealand journal of obstetrics & gynaecology. 2010;50(5):428-31.
5. Hannah ME, Ohlsson A, Wang E, Matlow A, Foster G, Willan AR. Maternal colonization with group B streptococcus and prelabour rupture of the membranes at term: The role of induction of labour. Am J Obstet Gynecol. 1997;177(4):780-5. (Level II).

Note:

These guidelines present recognised methods and techniques of clinical practice, based on published evidence, for consideration by, midwives, obstetricians/ gynaecologists and other relevant health professionals. The ultimate judgement regarding a particular clinical procedure or treatment plan must be made by the doctor or other attendant in the light of clinical data presented by the patient and the diagnostic and treatment options available. Departure from these protocols or guidelines should be fully documented in the patient's case notes at the time the relevant decision is taken and a “Declining recommended treatment” form signed by the patient and the clinician involved.

10. Glossary

The following definitions are relevant to this policy.

Term	Definition
AAU	Antenatal Assessment Unit – Outpatient clinic
GBS	Group B Streptococcus (GBS) status, may be referred to as “GBS positive”
IOL	Induction of Labour
PPROM	Preterm Pre-labour Rupture of Membranes
PROM	Premature Rupture of Membranes

11. Document owner

Enquiries relating to this guideline may be directed to:

Title: Consultant Obstetrician and Gynaecology
 Department: Obstetrician and Gynaecology
 Email: Abdoulaye.diarra@health.wa.gov.au

12. Review

This guideline will be reviewed and evaluated as required to ensure relevance and currency. At a minimum it will be reviewed within one (1) year after first issue and at least every three (3) years thereafter.

Version	Effective from	Effective to	Amendment(s)
3	17/05/2024	17/05/2027	4.7 Induction/augmentation with syntocinon infusion

13. Authorisation

This guideline has been authorised and issued by the [enter position title of the authoriser].

Authorised by	Tania Strickland – A/Director of Clinical Services
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